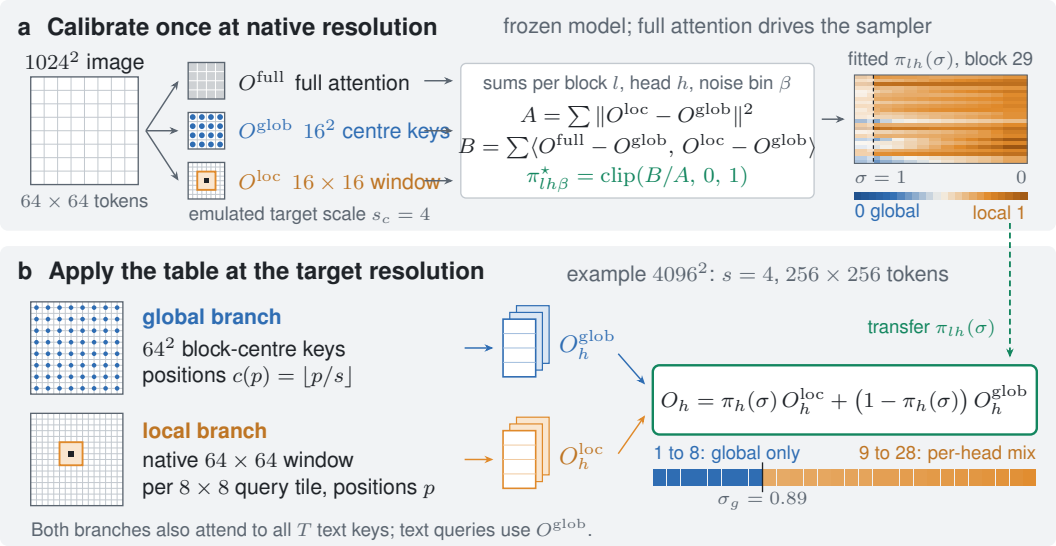
a Calibrate once at native resolution
frozen model; full attention drives the sampler
1024^2 image
64×64 tokens
O^{full} full attention
O^{glob} 16^2 centre keys
O^{loc} 16×16 window
emulated target scale $s_c = 4$
sums per block l, head h, noise bin β
$A = \sum \|O^{\text{loc}} - O^{\text{glob}}\|^2$
$B = \sum \langle O^{\text{full}} - O^{\text{glob}},\, O^{\text{loc}} - O^{\text{glob}} \rangle$
$\pi^{\star}_{lh\beta} = \text{clip}(B/A,\, 0,\, 1)$
fitted $\pi_{lh}(\sigma)$, block 29
$\sigma = 1$
0
0 global
local 1
b Apply the table at the target resolution
example 4096^2: $s = 4$, 256×256 tokens
global branch
64^2 block-centre keys
positions $c(p) = \lfloor p/s \rfloor$
local branch
native 64×64 window
per 8×8 query tile, positions p
O_h^{glob}
O_h^{loc}
transfer $\pi_{lh}(\sigma)$
$O_h = \pi_h(\sigma)\, O_h^{\text{loc}} + \big(1 - \pi_h(\sigma)\big)\, O_h^{\text{glob}}$
1 to 8: global only
9 to 28: per-head mix
$\sigma_g = 0.89$
Both branches also attend to all T text keys; text queries use O^{glob}.